HACKPSU 2026 / RESEARCH & PRODUCT STRATEGY

RESEARCH TWIN

AI-powered research discovery, paper understanding, and connected knowledge work

Product concept, system architecture, AI integrations, implementation roadmap, and source reconciliation

WHITE PAPER | WORKING VERSION 1.0

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Prepared from the team's original planning document, shared discussions, Research Twin product description, Goodnotes sketches, and selected official technical documentation.

Executive summary

Research Twin is a proposed AI-native research environment that helps students, early-stage researchers, and other academic readers move from an initial area of interest to credible source papers, practical comprehension, organized notes, and evidence-backed ideas for further research. It is not intended to be only a PDF summarizer or only a literature search engine. Its objective is to reduce the fragmentation between discovering literature, evaluating a paper, reading technical material, discussing it with AI, and turning what was learned into reusable research knowledge.

The product has two connected surfaces: (1) a web application serving as the primary research workspace, with project organization, literature discovery, paper previews, comparison, AI chat, notes, tabs, and an eventual knowledge graph; and (2) a Chrome extension that stays near the document being read to offer contextual explanations, prerequisite concepts, annotations, and conversational voice interaction. Both should operate on one shared paper model and one shared backend rather than building separate parsers or AI pipelines.

The research-discovery experience starts with a natural-language description of interests, goals, or a possible study. A bounded AI-guided workflow turns that description into search queries, retrieves verifiable scholarly records, expands through citations and related literature, and provides transparent relevance explanations. A data-first preview helps decide whether a paper is worth reading by surfacing methodology, sample or dataset details, reported findings, limitations, and access to supporting materials where available.

The reading experience begins with reliable PDF ingestion and source-aware analysis. Gemini is the team-named primary document-analysis service; ElevenLabs is the team-named voice layer. A context-aware voice agent can explain a selected paragraph or methodological decision and support follow-up dialogue, while the product retains a visible link between generated claims and the underlying paper. Notes and analyses belong to the same research project across both interfaces.

DECISION POSTURE This paper separates source-backed product requirements, technical recommendations, and unresolved decisions. A feature mentioned in sketches is not represented as already implemented or formally committed. The live GitHub codebase was not verified for this document.

The principal recommendation

Build one dependable vertical slice first: describe an interest → retrieve genuine papers → inspect a useful preview → open a PDF → ask for a contextual explanation in the extension → hear an ElevenLabs answer → save a note back to the same project. Advanced autonomous gap discovery, robust cross-paper contradiction detection, and an interactive graph can be phased in after this connected experience works.

Document status and interpretation

Label	Meaning
Stated direction	Appears in the supplied project discussions, product narrative, or design sketches.
Proposed design	Recommended architecture, UX refinement, or safeguard; requires team approval.
Open decision	Not yet settled in accessible project material.
Verified platform detail	Backed by the referenced provider documentation as reviewed October 2026.

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Section headings remain editable in the companion Word file. Fixed page references are omitted so the table of contents stays useful after Google Docs edits.

Project at a glance

Dimension	Current working definition
Product	Research Twin: AI-enabled scholarly discovery, understanding and connected notes
Clients	Research workspace web app + Chrome reading companion
Named AI services	Gemini document analysis; ElevenLabs voice; Grok role undecided
Core technical strategy	One shared backend, canonical paper identity, reusable analysis and evidence
Demo goal	Interest → real papers → preview → read → explain with voice → save note

01 Project context, problem and opportunity

1.1 The underlying problem

Reading and conducting research often involves a fragmented set of tasks. People identify a topic, search scholarly indexes, screen titles and abstracts, obtain full text, inspect methods and results, repeatedly look up prerequisite concepts, take notes in another application, and attempt to synthesize ideas across studies. The necessary information is distributed across browser tabs, PDFs, citation managers, text editors, and conversational AI tools. This creates context switching, duplicate work, and a risk that a useful observation is lost before it is connected to the broader project.

For readers new to a domain, the difficulty is not only the volume of papers. A paper may assume statistical, mathematical, methodological, or domain-specific knowledge that is not explained in its introduction. For people planning new work, it is also difficult to tell whether a research idea has already been studied, whether a cited result is applicable to their proposed setting, or whether an apparent limitation is a credible starting point for further investigation.

1.2 Research Twin's response

Research Twin aims to provide an evidence-centered layer across the research lifecycle. It helps users discover and filter literature, preview the substance of a paper before deep reading, understand unfamiliar material while retaining context, save and connect knowledge, and reason cautiously about disagreements and open questions. AI accelerates these activities; it does not replace reading primary sources or verifying scholarly claims.

1.3 Scope of the original team plan

The earlier team blueprint defined an intentionally small, modular Next.js application with one shared document-ingestion layer, typed API boundaries, an initial text/Markdown vertical slice, and a premium visual system [S1, S2]. At that stage it explicitly stated that the end-user problem had not been finalized. The later Research Twin description supplies the actual product mission [S3]. The Goodnotes sketches add concrete workspace and information-design ideas [S4]. Consequently, this paper updates—not discards—the earlier engineering direction.

1.4 Outcomes, not implementation claims

- **Faster literature screening:** Users can determine whether a paper may be relevant before investing in a full read.
- **Contextual comprehension:** Explanations remain tied to the paper, selected passage, and necessary prerequisite concepts.
- **Connected knowledge:** Notes, papers, projects, and citations stay accessible across the web app and extension.
- **Evidence-backed synthesis:** Possible gaps, limitations, and contradictions are explained using traceable sources.
- **A credible hackathon demo:** A small but complete connected workflow works reliably under real user interaction.

02 Users, jobs-to-be-done and product principles

2.1 Primary and secondary users

The precise target audience was not settled in the earliest planning document. Based on the Research Twin concept, a sensible initial focus is university students, research newcomers, and researchers exploring unfamiliar subfields. More advanced users may benefit from cross-paper synthesis, citation exploration, and structured evaluation, but should not be assumed to be the initial product-market fit.

User	Research job	Most important capability
Student entering a field	Understand technical papers without losing reading context	Prerequisites, clear explanations, source-aware AI Q&A
Early-stage researcher	Move from broad interests to a defensible literature set	Topic-driven discovery, evidence filtering, save-and-compare
Project-based reader	Read several papers and retain reusable knowledge	Project library, notes, tags, links, tabs
Research proposer	Investigate a potential method, limitation or unanswered question	Data/method preview, critique, cautious gap suggestions

2.2 Jobs-to-be-done

- "Help me find what researchers have already published about an idea I can describe in ordinary language."
- "Before I read the whole paper, show me what data it uses, how the study was conducted, and whether its evidence matches my needs."
- "I do not understand this paragraph or equation; explain it, tell me the prerequisites, and let me explore one of them without leaving the document."
- "Let me ask the paper questions by speaking, and show where the answers come from."
- "Save my notes and link them to this paper, my project, and other relevant findings."
- "Help me evaluate a proposed claim against several papers and investigate what might still be unknown."

2.3 Product principles

01 **One connected workflow.** The research project, PDF, note and AI conversation should not feel like separate products.

02 **Evidence before eloquence.** Users must be able to find the source for material claims; unverified inference must be labeled.

03 **Progressive disclosure.** Give a fast preview first; reveal methodology, tables, citation paths and caveats on demand.

04 **Researcher control.** Users choose sources, save notes intentionally, confirm selections and can correct the model.

05 **Readable, quiet interface.** Typography, density, tabs and restrained motion support sustained reading.

06 **Hackathon realism.** Favor a working end-to-end system and stable contracts over premature infrastructure or agent complexity.

03 Unified end-to-end experience

3.1 The core research lifecycle

Stage	User action	Product response
Discover	Describe topic, area and goals	Suggest refinements; retrieve identifiable papers; explain relevance
Screen	Inspect result cards and data previews	Show methods, datasets/samples, outcomes, limitations, PDF access
Understand	Open PDF and select confusing content	Provide grounded explanations, concepts and voice tutoring

Connect	Save papers, notes and tags	Build a per-project research library and navigable links
Evaluate	Compare and critique selected research	Show differences, possible contradictions and tentative gaps
Extend	Plan a new question or improvement	Turn evidence-backed limitations into testable research directions

3.2 Example research journey

A user enters: “I am interested in personalized AI tutoring and whether it improves long-term learning.” Research Twin asks or infers minimal useful facets—population, setting, outcome, preferred publication window—and searches a scholarly index. It presents a small, real list of papers, not invented citations. Each card explains why the result appears relevant and distinguishes an abstract-only view from a fully analyzed PDF.

The user previews a study’s sample, experimental design, measures, reported effect or findings, time horizon, and limitations. They save it to an “AI Education” project and open it in the research workspace. During reading, a difficult paragraph can be selected. The extension offers a short explanation, prerequisite links, and “Explain with Voice.” The voice assistant answers the question using the selected passage and related paper sections. The user saves the explanation as a note linked to the paper.

After reviewing additional papers, the user asks whether the results support long-term retention. Research Twin separates studies by outcome measures and evaluation periods, flags differing conditions, and proposes questions worth investigating. The result is a better-defined research direction rather than an AI claim that a verified knowledge gap exists.

3.3 The relationship between the two surfaces

Web application — the primary environment for initiating research, curating projects, reviewing PDFs, comparing studies, managing linked notes, viewing multiple tabs, and ultimately navigating research graphs.

Chrome extension — the contextual companion alongside web content and research PDFs: quick explanations, prerequisites, short notes, paper actions and voice access without abandoning the reading task.

Shared services — one paper identity, one parsing/analysis pipeline, one research index adapter, one note store, and one source-reference scheme accessible to both clients.

04 Web app: discovery and research workspace

4.1 Topic-first onboarding and project creation

The first interaction should accept an imprecise description such as a field of interest, a possible problem, an intended methodology, or a desired population. The system can help refine this into search dimensions while leaving control with the user: topic, relevant subtopics, exclusions, recency window, discipline, evidence type, and optional seed papers. Save these dimensions to a research project so discovery is reproducible rather than a one-off chat transcript.

4.2 Literature-discovery assistant

01 **Clarify the intent.** Convert natural language into an explicit research question or a set of search facets; record assumptions.

02 **Query reliable scholarly sources.** Use an external bibliographic API rather than asking a language model to generate citations.

03 **Normalize and deduplicate.** Merge records by DOI, platform identifiers and conservative title/author matching.

04 **Score and explain.** Rank by topical fit, research methodology, accessible text, publication date and user preferences; do not assume recent means better.

05 **Expand citation neighborhoods.** Use references, cited-by and related-work relationships when supported by the selected source.

06 **Show real links and uncertainty.** Indicate missing abstracts, unavailable PDFs, missing metadata or uncertain relevance.

07 **Save the search session.** Preserve queries, filters, retrieved records, and why items were recommended.

The Goodnotes notes specifically name Web of Science for metadata and citation discovery followed by PDF retrieval [S4, p. 2]. That is a genuine team suggestion, not a confirmed integration. Web of Science's Starter API has access-tier limits [R1]. OpenAlex and Semantic Scholar are viable alternatives or complementary providers [R2, R3]. Provider selection and available API credentials remain open.

4.3 Paper preview and data-first screening

The user requested a data-first view to determine whether a paper contains the kind of evidence they need. This should be prominent, not buried beneath a long general summary. A preview can display title, authors, year, venue, DOI, abstract, research question, domain, design, dataset or study sample, variables, measurements, evaluation methods, principal findings, effect estimates when reported, limitations, availability of data/code, and access links.

Critical distinction: the numbers and tables reported in a paper are not necessarily the underlying raw dataset. A preview should separately label "reported evidence," "linked dataset," and "raw data not available." It should never hallucinate a dataset size, numerical effect, p-value, or access URL. When fields are unavailable, say "not reported," "not retrieved," or "not yet analyzed," as appropriate.

4.4 Paper evaluation, feedback and improvements

Users want to assess how an existing study might be improved. The system can provide structured critiques of internal validity, external validity, sample composition, methodological choices, benchmark suitability, replicability, uncontrolled confounders, evaluation duration, missing ablations, limitations acknowledged by the authors, and reproducibility artifacts. Each suggestion should be phrased as a consideration: "A longer follow-up might test whether the effect persists," rather than "The authors' conclusion is wrong."

Critiques must distinguish (a) limitations explicitly disclosed by the paper, (b) reasoned inferences based on evidence, and (c) unanswered questions requiring external verification. Present these as different evidence classes.

4.5 Multi-paper comparisons and apparent contradictions

A comparison workspace should align studies along dimensions the user actually cares about: population, intervention, data source, metrics, study duration, baseline, limitations, reported outcomes and uncertainty. A "contradiction" requires comparable claims under sufficiently comparable conditions; disagreements may be due to differing definitions, cohorts, data preprocessing, tasks or measurement choices. Research Twin should show these alternative explanations before claiming the literature conflicts.

4.6 Research gap exploration

Gap finding is a hypothesis-generation capability. Useful signals include underrepresented populations, inconsistent results, short evaluation horizons, missing comparison baselines, unexplored combinations of methods, limited external validation, and recurring author-reported limitations. The product should display the papers considered, evidence snippets, limitations of the search corpus, confidence and suggested verification queries. It must never describe a gap as definitely novel merely because it is absent from a small retrieval set.

4.7 Research library, notes and graph

The Goodnotes sketches imply an Obsidian-like working environment: left navigation containing projects, folders, papers and notes; multiple open tabs; split-screen panes; tags or wiki-style note links; and graphs connecting notes, papers and potentially entire projects [S4, pp. 1, 3–4]. The user explicitly wants the ability to preserve and revisit research found during early exploration [S3].

- **Projects and folders:** Organize a literature review, class topic or planned study; support importing papers and notes.
- **Workspace tabs:** Keep an article, project overview, note and AI conversation open in parallel.
- **Split view:** Place a PDF or paper analysis alongside notes, comparisons or the graph.
- **Annotations:** Store highlighted passages and user-authored notes with source anchors.
- **Tags and backlinks:** Link concepts, notes, papers and projects explicitly; show backlink context.
- **Knowledge graph:** Distinguish real citation edges, user-created links and AI-suggested relationships with visible edge types.
- **Focused dashboard:** Resume recent projects and unfinished reading; avoid a generic metric-heavy landing page.

05 Chrome extension: contextual research companion

5.1 Intended experience

The extension opens as a persistent side panel beside a source document. The user can upload a PDF, use a supported page, or open a paper in a compatible viewer. The extension identifies the active paper when possible, retrieves its analysis from the shared backend, and exposes clear actions: Overview, Prerequisites, Explain Selection, Notes, and Voice. It should not attempt to reimplement the website's full project dashboard.

5.2 Primary reading actions

Action	Behavior	Important constraint
Analyze current paper	Display question, methods, key results, limits, relevant figures	Source-aware and cached when possible
Check prerequisites	Extract assumed concepts; order by dependency or difficulty	Explain in context, not a generic syllabus
Explore a prerequisite	One click opens short explanation + example + return link	Retain reading position

Explain selection	Use exact user-selected passage with surrounding sections	Show quoted anchor and cited pages
Take notes	Capture note, source location and project destination	User controls save behavior
Ask with voice	Start contextual ElevenLabs conversation	Microphone permission and visible stop
Check claims	Inspect how a passage supports a conclusion; show caveats	Separate verification from speculation

5.3 PDF capture and browser constraints

The team's early Chrome-extension outline assumes "PDF opened → extract PDF content → analyze using Gemini" [S3]. This is the right product intention but requires careful engineering. Chrome's extension content scripts primarily operate on accessible web-page DOMs; the built-in browser PDF viewer should not be treated as an ordinary page whose text can always be read or selected. Availability may vary by viewer, file URL, browser permissions and hosting. A reliable MVP needs a controlled path: upload a PDF directly, open it in a viewer the team owns, or process an explicitly supplied accessible PDF URL after validation. Ordinary HTML article pages can be supported separately. Test the exact target reading environment early [R4, R5].

5.4 How contextual retrieval should work

01 Resolve the active paper from a stable ID, uploaded hash, known DOI or matching metadata; allow manual correction when ambiguous.

02 Collect the selection and a small surrounding context window, along with section and page references if supported.

03 Send a structured request to the shared API rather than embedding AI provider credentials in the extension.

04 Retrieve the necessary normalized paper passages and previously computed analysis, then request an explanation.

05 Return explanation, prerequisites and an evidence anchor; let the user jump back to the relevant location.

06 Persist only deliberately saved notes; make transient selection data ephemeral where feasible.

5.5 Extension implementation boundary

Use Chrome Manifest V3 and its Side Panel API for the docked companion [R4]. Keep the extension bundle compact: side-panel UI, permissions, optional content scripts for supported pages, background/service-worker messaging, and a small typed API client. Most AI logic belongs in the shared server. Minimize broad host permissions, provide clear file/access controls, and offer a graceful fallback when no paper can be extracted.

06 Paper intelligence and scholarly evidence

6.1 The ingestion pipeline

01 Accept an upload, authorized URL or known accessible full-text source; validate MIME type, byte size and access eligibility.

02 Identify paper metadata (including DOI and title where possible) and calculate a stable hash to avoid duplicate processing.

03 Extract native text and preserve page/section boundaries; handle tables and figures through a multimodal PDF path when useful.

04 Normalize extracted content into a paper-and-sections representation, keeping provenance to the source page.

05 Run structured extraction to detect research objectives, methodologies, datasets, samples, methods, results, limitations and concepts.

06 Validate required fields, record what could not be extracted, and cache results with analysis version and model identification.

07 Serve consistent representations to web workspace, extension, comparison, and voice experiences.

The initial shared ingestion proof-of-concept specified text and Markdown parsing. For this product, PDF support is a P0 requirement rather than a distant future option. Gemini documentation supports native PDF understanding including diagrams, charts and tables, but practical extraction quality, cost, and source localization must still be tested on target papers [R6].

6.2 Proposed analysis output schema

Category	Suggested structured fields
Identity	Title, author list, venue, year, DOI, canonical URLs, license/access
Study framing	Research question, hypothesis, scope, domain, population
Methods	Design, comparison groups, measurement approach, statistical methods
Evidence	Sample/dataset details, variables, reported numbers, tables/figure pointers
Results	Main findings, metrics, uncertainty, reported confidence/significance
Assessment	Author-stated limitations, inferred questions, replicability resources
Reading support	Sections, summary, prerequisites, glossary, source anchors
Provenance	Page/section/span, source URL, extraction status and analysis version

6.3 Evidence anchor design

Every material machine-generated research claim should ideally reference one or more text spans, sections, figures or tables within a known paper. A structured anchor could include paperId, page, sectionHeading, selectedText or quote, and a retrieval identifier. Page-number fidelity must be verified because PDF page indices, printed page numbers and rendered viewer positions can disagree. If exact highlighting is not available, show the best section/page reference and label the limitation.

6.4 Prerequisite map

A paper-aware prerequisite engine identifies concepts needed to read a specific document. It should separate fundamental background (e.g., probability), domain concepts (e.g., attention mechanisms), and paper-specific terms. Each concept offers a concise explanation, one accessible example, why it matters in the current

passage, and optional further reading. A concept graph is proposed as an implementation detail; full prerequisite personalization is longer-term.

6.5 Quality checks and human review

- Never manufacture citations, DOI values, dataset links or quantitative results.
- Expose coverage: abstract-only, full-PDF available, table evidence retrieved, or citation-only metadata.
- When a field is absent, distinguish “not stated in source” from “not processed yet.”
- Treat PDF extraction errors and reference mismatch as visible failures, not silent success.
- Allow the user to open original excerpts, adjust the interpretation, and report inaccurate analyses.
- Treat generated explanations as reading aids rather than proof that a paper is methodologically sound.

07 Agents, models and third-party integrations

7.1 Roles for technologies named by the team

Technology	Proposed function	Source / status
Next.js / React / TypeScript / Tailwind	Web application, API layer, typed contracts, visual UI	Original technical blueprint [S1]
Gemini	PDF/multimodal analysis, structured paper extraction, Q&A	User-named preferred analysis service [S3]
ElevenLabs	Real-time voice agent and spoken explanations	User-named voice integration [S3]
Grok	Optional second-model experiments or specialized review	Named but no required role defined [S3]
Cursor	Developer productivity/IDE, not a runtime service	Named development tool [S3]
Web of Science	Literature records, citations and metadata	Goodnotes proposal [S4, p. 2]
OpenAlex / Semantic Scholar	Alternate or complementary discovery providers	Architecture recommendation [R2, R3]
Vitest / GitHub Actions	Tests and automated checks	Original blueprint [S1]

7.2 Research discovery as a bounded agent

The user asked for an AI agent that helps discover existing work before starting research. A reliable initial agent is a constrained workflow, not an unrestricted autonomous crawler. It can iteratively reformulate a question, call a supported scholarly API, inspect metadata, follow references or related works, stop after a capped number of steps, deduplicate outputs, and write a summary grounded in returned records. Keep a human-visible search trail: date searched, provider, filters, retrieved count and cited source IDs.

An autonomous “research is complete” claim would be unjustified. The product should describe its search coverage and invite users to continue exploring.

7.3 Tool orchestration and dependency sequence

01 Search scholarly index through one provider adapter.

02 Normalize metadata and rank results using deterministic signals plus AI-relevance explanation.

03 Resolve open-access/full-text links when available and permitted.

04 Send a chosen PDF to the shared ingestion and Gemini analysis pipeline.

05 Store structured results and source anchors.

06 Expose retrieval tools to text chat and the ElevenLabs agent.

07 Allow deliberate saving of notes and knowledge links.

7.4 External-provider decisions

Web of Science is listed by the team, but access may depend on institutional subscriptions and rate limits; its Starter API free trial advertises 50 requests per day with limited metadata compared with subscription tiers [R1]. OpenAlex supports works search and cited-reference traversal [R2]. Semantic Scholar documents its Academic Graph and Recommendations APIs [R3]. To keep architecture flexible, implement one adapter interface with provider-specific mapping and test at least one source under actual hackathon credentials.

The presence of three AI-related brands does not establish that all three must be involved in every query. Fewer integrations with clear ownership reduce latency, token costs and contradictory outputs.

08 Shared technical architecture and data model

8.1 One repository, separate clients, shared services

The original team preferred a single straightforward repository, reusable services, typed APIs, and minimal infrastructure [S1, S2]. Preserve that preference. The main Next.js application can provide both the website and server routes. A separately bundled browser extension can live in an extension directory inside the same repository. It is a distinct client, not simply another Next.js page. Both call the same server-side services.

RECOMMENDED REQUEST PATH Web app or Chrome side panel → typed application API → research discovery / ingestion / analysis / evidence retrieval / voice-session services → shared paper and project store. Only server-controlled routes handle private external API keys.

8.2 Proposed minimal repository outline

```
src/app/                    Web pages, layouts and API routes
src/components/             Shared UI components
src/features/               Research-, paper-, workspace-specific features
src/lib/ingestion/          PDF and document processing
src/lib/research/           Scholarly search and citation adapters
src/lib/analysis/           Gemini extraction, grounded Q&A, evaluation
src/lib/voice/              ElevenLabs session/tool integration
src/lib/storage/            Paper, project and note persistence
src/types/                  Cross-client types and API contracts
extension/                  Side panel, manifest and extension integration
docs/                       Product, architecture, design and contribution notes
```

The directory list is a target organization, not a verified inventory of the current GitHub repository. Folders should be added only when they hold useful implementation.

8.3 Canonical concepts

Entity	Responsibility	Important link
ResearchProject	User-defined topic, scope, saved library and project state	Contains PaperMembership and Note
Paper	Canonical bibliographic identity, sources and metadata	May exist without an analyzed PDF
DocumentVersion	Fetched/uploaded file, checksum, extraction progress	Belongs to one Paper

PaperSection / EvidenceSpan	Page-aware normalized evidence	References DocumentVersion
PaperAnalysis	Versioned structured interpretation	References evidence spans
Note / Annotation	User-authored knowledge and selected-source anchor	Links to Paper and project
Relationship	Citation, note link, inferred topical relation	Typed and attributable
Conversation	Contextual text/voice exchange	Links to active paper/project

8.4 Suggested API boundaries

Route family	Illustrative operations
/api/research/search	Convert topic and facets to scholarly search results
/api/papers	Create, lookup, resolve bibliography and fetch metadata
/api/papers/ingest	Upload/fetch document and begin normalized extraction
/api/papers/{id}/analysis	Read cached result or request structured analysis
/api/papers/{id}/ask	Evidence-aware answers about paper or selection
/api/projects	Create/list projects and project-paper memberships
/api/notes	Create/update anchored notes
/api/voice/session	Authorize/initiate agent conversation with paper context

These are suggested endpoint families; final route naming, request bodies, auth and streaming protocols should be established through short shared TypeScript contracts before separate contributors start coding.

8.5 Processing states and errors

Document processing should expose at least queued, processing, completed and failed states. Responses should use consistent error shapes and machine-readable codes for bad format, oversized upload, missing access, provider quota, PDF parse failure, no supporting evidence, and expired voice session. Idempotent ingestion and source hashing reduce duplicate work and wasted API calls.

8.6 Persistence and authentication

The original blueprint left storage and sign-in open. A cross-surface library of saved papers and notes requires a persistence decision: local-only state can demo basic workflows but cannot automatically sync between browsers and devices; shared server-side persistence creates account/identity and access-control needs. For a hackathon, choose the least complex path that supports the demonstrated flow, define ownership of uploaded files and notes, and avoid silently relying on an unimplemented user identity system.

09 ElevenLabs voice integration specification

9.1 The intended experience

The ElevenLabs integration is not merely reading a summary aloud. It is a contextual spoken research tutor that accepts a user question about the current paper or selected passage, maintains short conversational

continuity, and gets grounded information from Research Twin's own evidence service. The voice interface complements text chat rather than competing with it.

9.2 Voice use cases

- Explain the selected paragraph in simpler terms while preserving the meaning of key terminology.
- Break down a method, equation, chart or experimental choice into understandable steps.
- Explain a prerequisite and return to the original paragraph.
- Answer follow-up questions about the current paper's aims, evidence, and limitations.
- Compare findings with a second saved paper only after multi-paper retrieval is working.
- Show the answer transcript and supporting source locations in the side panel.

9.3 Recommended service interaction

01 A user explicitly starts voice mode and grants microphone access.

02 The client obtains an authorized, short-lived ElevenLabs session credential from a trusted backend route; secrets remain server-side.

03 The voice agent receives an initial scoped context: paper ID, selected passage, section, optional project ID and permitted tools.

04 For factual paper questions, the agent calls a backend retrieval tool that returns compact, cited evidence and a constrained answer.

05 ElevenLabs turns that answer into conversational audio; the UI displays the corresponding text and cited sources.

06 When the user changes the selection or paper, update the active context without leaking unrelated project material.

07 The user can mute/stop; the app ends the session and applies its selected retention policy.

ElevenLabs' React SDK documents conversation sessions, client tools, message callbacks, and contextual updates [R7]. Its agent authentication documentation recommends server-generated signed URLs for protected client connections, avoiding exposure of the API key in browser code [R8]. Exact credential and session mechanisms should follow the chosen SDK version and account configuration.

9.4 Proposed voice context contract

At minimum, a voice session should know paperId, optional selectedText, optional section/page anchor, optional projectId, an evidence-access scope, and a session identity. Conversation history should not be treated as the authoritative source of paper facts. The backend must continue to retrieve and verify references on demand.

9.5 Definition of done for the voice MVP

- Starts and stops reliably in the supported web/extension context; microphone status is visible.
- Correctly recognizes which paper and selection the user is discussing.
- Answers three prepared questions with explanations supported by visible PDF sections.
- Gracefully handles missing paper context, quota/network failure, permissions denied and user interruption.
- Does not expose API keys in client code or source control.
- Can save a chosen explanation as a note if the shared note API is available.

10 Interaction design and visual direction

10.1 Design thesis

The team explicitly wants a premium, non-generic interface: disciplined typography, restrained color, consistent spacing, responsive behavior, and purposeful motion [S1, S2]. The Goodnotes sketches make the reference style tangible: Obsidian-like compact side navigation, file/project hierarchy, workspace tabs, split screens, and a graph panel [S4, pp. 1, 3–4]. The sketches should inform information architecture and interaction patterns; they are not finalized pixel-perfect designs.

10.2 Suggested navigation anatomy

- **Left rail/sidebar:** Project switcher, folders, papers, notes, search, saved views and graph entry.
- **Main editor/reader:** Tabbed area containing a PDF, paper analysis, project overview, note or comparison.
- **Optional split pane:** Paper alongside notes, AI chat, comparison or knowledge graph.
- **Context assistant:** Small AI panel or action menu that understands selected paper/passage.
- **Extension side panel:** Condensed version of contextual actions, not the entire dashboard.
- **Status surfaces:** Clear indications for document processing, source coverage, active voice state and unsaved notes.

10.3 Typography, motion and density

Use a small type scale with strong hierarchy. Fonts such as Geist or Inter were mentioned in the original blueprint; Inter is a practical choice for this paper. Default to neutral surfaces and one restrained accent. Avoid ornamental gradients, excessive glass effects, oversized cards and arbitrary animations. Papers and notes need denser information display than a marketing homepage, but there must still be adequate line length, spacing, keyboard focus and accessibility.

The shared design resources include UIBall loading animation references and a spreadsheet titled “The People’s Design Library” [S5]. These are inspiration links rather than an agreed design system. Motion should communicate useful states—searching, ingesting, generating analysis and establishing voice—not merely decorate an interface.

10.4 Interaction examples

Moment	Good interaction	Avoid
First search	Short interest prompt plus optional advanced facets	Demanding a long form up front
Paper card	Methods/results preview with source/coverage labels	Generic abstract rewritten as certainty
Selection	Small menu: Explain / Prerequisites / Note / Voice	Replacing the paper with a full-screen chat
Prerequisite	Inline concept, example, and return link	Opening a large unrelated lesson
Voice	Listening/speaking, transcript, stop, evidence	Audio without readable provenance
Knowledge graph	Clear edge types and project scoping	Undifferentiated decorative network

11 Safety, scholarly integrity and risk management

11.1 Academic accuracy and research integrity

Research Twin should be designed to assist scholarly judgment, not impersonate it. LLM responses can be fluent despite retrieval failures or incorrect interpretations. High-value safeguards are source-grounded claims, visible confidence limits, coverage labeling, and explicit separation of author statements from model inference. The platform should be clear that summaries and suggestions are aids to review, not peer review or proof of novelty.

11.2 Data, privacy and rights

- Do not upload, redistribute or cache paywalled or restricted full text without the relevant rights or authorization.
- Protect API keys on the server, restrict CORS, validate extension origin and validate external URLs to reduce SSRF risk.
- Request only the browser permissions necessary for the supported reading experience.
- Provide deletion/retention controls for user-uploaded PDFs, notes and voice transcripts if they are stored.
- Do not reveal project-private notes or documents across sessions or between users.
- Make microphone use opt-in and visually clear; never start voice recording silently.
- Sanitize untrusted webpage and PDF content; treat embedded instructions in documents as data, not as commands to the agent.

11.3 Known product and engineering risks

Risk	Why it matters	Mitigation
Incorrect citations	Can undermine user trust and research integrity	Retrieve real records; validate IDs/DOIs; link evidence
Poor PDF extraction	Charts, tables and page anchors may be lost	Multimodal path, coverage states, representative test corpus
Overclaimed research gaps	“No paper exists” cannot follow from limited search	Qualified language, saved search trail, user verification
False contradictions	Results may not be methodologically comparable	Comparison matrix with cohort/metric/context caveats
Extension access limits	Built-in PDF viewer may not expose content	Controlled upload/viewer fallback; early browser test
Third-party quotas/costs	Search/analysis/voice calls may be capped	Caching, rate limits, narrow demo corpus, graceful errors
Feature sprawl	Many half-finished capabilities weaken demo	Freeze P0 workflow first; isolate stretch features

12 Delivery strategy, ownership and testing

12.1 Recommended milestone sequence

Phase	Goal	Deliverable
0 — Align	Commit to actual product requirements and contracts	PRODUCT.md, ARCHITECTURE.md, ownership map, test corpus
1 — Foundation	Prove a shared ingestion/analyze route	PDF upload/parse, Gemini structured analysis, consistent errors
2 — Discovery	Retrieve genuine scholarly records	Search provider adapter, dedupe, paper cards/preview
3 — Reading	Connect extension to the same paper API	Side panel, selected-text analysis, prerequisites, notes stub
4 — Voice	Answer a paper question by conversation	ElevenLabs contextual session + evidence-backed response
5 — Finish	Cross-surface polish and robust demo	Saving project state, integration tests, failure handling
Stretch	Increase research synthesis or workspace depth	Graph, richer tabs/split view, multi-paper contradictions

12.2 Proposed workstreams, without assuming individual assignments

- **Shared paper engine:** file validation, canonical paper identity, PDF extraction, normalization, evidence positions, caching.
- **Research discovery:** scholarly provider adapter, AI-assisted query construction, related/citation traversal, result UI.
- **Web workspace and design:** project dashboard, paper preview, tab layout, note experience, typography and interaction consistency.
- **Chrome extension:** manifest, side panel, supported reading contexts, selected-text capture, typed API client.
- **Voice integration:** ElevenLabs session orchestration, paper context, transcripts, listening states and speaker-facing citations.
- **Integration and QA:** end-to-end demo tests, quota handling, documentation, release/deployment readiness.

12.3 Team coordination rules inherited from the blueprint

Keep one repository, use short-lived feature branches, merge small pull requests regularly, document shared types before changing their semantics, and keep implementation notes concise. The shared TypeScript contracts should be reviewed together; feature UI can be developed against mock JSON while server processing is built. Avoid a monorepo migration, separate microservices or needless event infrastructure without demonstrated requirements.

12.4 Minimum test matrix

Level	Test examples
Unit	Supported/unsupported uploads, metadata normalizer, source anchor serializer, API schemas

Provider integration	Valid/empty search results, quota errors, missing DOI, inaccessible full text
Analysis quality	Methods/data/limitations against manually checked sample papers
Cross-client	Same paper identity and analysis shown in website and extension
Voice	Paper context, selected-text explanation, permission denial, session stop
End-to-end	Research interest → paper → PDF → explanation → saved note
Accessibility	Keyboard focus, readable contrast, side-panel navigation and controls

13 Demonstration and success criteria

13.1 Demo story

Choose a research topic with a small, permitted set of authentic papers that are easy to verify. The demonstration starts with a human-language research interest, shows a real literature search and compelling method/data preview, opens a chosen paper, uses the extension to explain a difficult passage and prerequisite, asks the ElevenLabs agent a follow-up question, and saves the result in the web workspace. If prepared, show an additional paper comparison or a cautious candidate research gap as a bonus.

13.2 Suggested evaluation criteria

- The same saved paper is identifiable across discovery, analysis, extension and library.
- Every displayed bibliographic citation resolves to a real source.
- Key analysis fields can be located in the sample source document or are marked uncertain.
- A new user can preview data/methods without reading the entire paper.
- The extension returns a correct, contextual explanation for a selected span.
- Voice state is transparent, follow-up dialogue works, and the agent uses the current paper context.
- A note can be created and retrieved in the intended research project.
- The demo survives common failure paths or clearly communicates the problem.

13.3 Long-term product measures (not claims of current performance)

Potential measures include time to identify a relevant paper, precision of saved-paper recommendations, proportion of AI claims with usable source anchors, successful paper ingestion rate, user-rated comprehension after an explanation, note retention/retrieval, cross-client task completion, and voice-task success. These are future evaluation dimensions; the project has not supplied baseline performance measurements or deployed-user evidence.

14 Open decisions and reconciliation

14.1 What came from which source

Theme	Original team blueprint	Later product / Goodnotes detail
Product purpose	Generic document-aware application; final problem open	Research Twin research discovery and reading assistant
Experiences	Two site-like areas in one app	Website + real Chrome extension

Files	Initial TXT/Markdown slice; PDF possible later	Research PDFs must be first-class
Discovery	Not specified	Topic-driven agent; Web of Science and citation expansion
Screening	Not specified	Data-first methods, sample/results, limitations
Reading	Shared parsed content	Selected-text explanation and prerequisite deep dives
Knowledge	No storage contract	Projects, note folders, tabs, split view, links, graph
AI / voice	Future integrations undecided	Gemini analysis; ElevenLabs dialogue; Grok mentioned
Engineering / design	Minimal shared Next.js, typed API, premium UI	Still aligned; stronger product detail required

14.2 Differences that require explicit decisions

01 **Paper source access.** Determine which scholarly API keys and institutional resources the team actually has.

02 **PDF reader approach.** Pick a reliable first-class PDF view/upload route and supported extension contexts.

03 **Data persistence.** Decide whether notes and projects sync through server storage or are initially local.

04 **Scope of the graph.** Confirm whether it is a demonstrable milestone or just a longer-term UX reference.

05 **Voice knowledge owner.** Define which service answers grounded factual questions and how ElevenLabs consumes it.

06 **Model policy.** Confirm whether Grok is a required runtime integration or only a tool being explored.

07 **Auth and deployment.** Agree on enough access control and hosting to support the demonstration.

08 **Priorities and owners.** Translate P0/P1/P2 into named assignments and a common integration test.

14.3 What should be treated as the current source of truth

The product's conceptual source of truth should be a concise, agreed PRODUCT.md based on the Research Twin vision and the Goodnotes information architecture. The earlier engineering blueprint should be rewritten as ARCHITECTURE.md with the actual website/extension distinction and PDF-centered contracts. The present white paper is a reconciled proposal—not a substitute for team decisions or proof of finished implementation.

Appendix A Representative API contracts

A.1 Canonical paper record — illustrative

```
type Paper = {
  id: string;
  title: string;
  authors: string[];
  year?: number;
  doi?: string;
  sourceUrl?: string;
  pdfUrl?: string;
  abstract?: string;
  ingestStatus: "not_started" | "processing" | "ready" | "failed";
};
```

A.2 Evidence-aware analysis — illustrative

```
type EvidenceAnchor = {
  paperId: string;
  documentVersionId?: string;
  pageIndex?: number; // viewer page, if verified
  section?: string;
  quotation?: string;
};

type SupportedFinding = {
  claim: string;
  evidence: EvidenceAnchor[];
  kind: "author_reported" | "model_inference" | "unknown";
};
```

A.3 Voice context — illustrative

```
type VoiceContext = {
  paperId: string;
  projectId?: string;
  selectedText?: string;
  pageIndex?: number;
  section?: string;
};
```

These interfaces are proposals designed to start cross-team discussion, not final production contracts. The user asked for simple architecture; interfaces should remain minimal until tested against real product operations.

Appendix B Source register and external references

B.1 Project materials supplied or surfaced

[S1] HackPSU 2026 — Project Concept & Technical Blueprint, v0.1, October 24, 2026. Supplied as a LaTeX-formatted Google Docs export; stated goals, stack and development phases.

[S2] Shared Hack PSU 2026 project conversations, October 24, 2026. Architecture planning, minimal repo, shared ingestion, GitHub/Codex handoff, premium visual direction.

[S3] Research Twin product narrative supplied in conversation, October 24, 2026. Discovery agent, papers, data-first preview, paper critiques/gaps, Chrome extension, Gemini, ElevenLabs and Grok.

[S4] Research Hack.pdf, four-page Goodnotes export supplied in conversation. p. 1 dashboard mock-up; p. 2 Web of Science/citation idea; p. 3 Obsidian-like tabs/split views/note links/graphs; p. 4 component reference.

[S5] Team UI reference links: https://uiball.com/ldrs/ and the shared “The People’s Design Library” Google Sheet. Links alone do not define approved components.

[S6] Repository shared by user: https://github.com/builtbybayn/hack-26 . Current file contents and latest branches were not independently verified in this document.

B.2 Official technical references

[R1] Clarivate, Web of Science Starter API.

https://developer.clarivate.com/apis/wos-starter

[R2] OpenAlex, API reference and citation-search recipes.

https://help.openalex.org/api/ | https://help.openalex.org/how-to/api-recipes/

[R3] Semantic Scholar, Academic Graph and Recommendations API documentation.

https://api.semanticscholar.org/api-docs/

[R4] Chrome Developers, Side Panel API.

https://developer.chrome.com/docs/extensions/reference/api/sidePanel

[R5] Chrome Developers, Content scripts: capabilities and limitations.

https://developer.chrome.com/docs/extensions/develop/concepts/content-scripts

[R6] Google AI for Developers, Gemini document understanding.

https://ai.google.dev/gemini-api/docs/document-processing

[R7] ElevenLabs, React SDK for conversational agents.

https://elevenlabs.io/docs/eleven-agents/libraries/react

[R8] ElevenLabs, Agent authentication and signed URLs.

https://elevenlabs.io/docs/eleven-agents/customization/authentication

B.3 Interpretive caveats

This white paper is a design and engineering proposal, not a scientific efficacy study, independent test report, or representation that the described product has already shipped. Source-backed requirements, reasoned design proposals and undecided choices are deliberately separated. The Goodnotes sketches are early concepts; ambiguous handwritten fragments have not been expanded into invented requirements. No live repository inspection was successful, so no current implementation status is asserted.

All external platform guidance is subject to change. Before implementation, re-check available product tiers, API versions, permissions, provider quotas, terms of use and SDK instructions in official documentation.